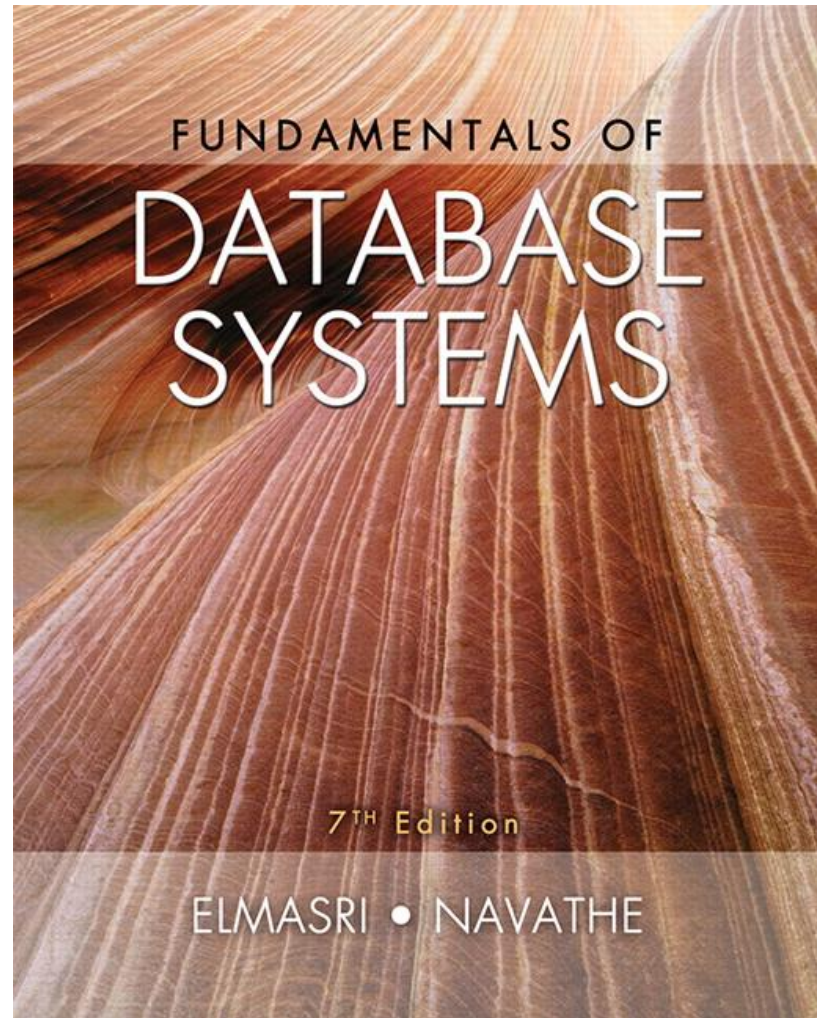




CHAPTER 20

Introduction to Transaction Processing Concepts and Theory

Introduction

- Transaction
 - Describes local unit of database processing
- Transaction processing systems
 - Systems with large databases and hundreds of concurrent users
 - Require high availability and fast response time

20.1 Introduction to Transaction Processing

- Single-user DBMS
 - At most one user at a time can use the system
 - Example: home computer
- Multiuser DBMS
 - Many users can access the system (database) concurrently
 - Example: airline reservations system

Introduction to Transaction Processing (cont'd.)

■ Multiprogramming

- Allows operating system to execute multiple processes concurrently
- Executes commands from one process, then suspends that process and executes commands from another process, etc.

Introduction to Transaction Processing (cont'd.)

- Interleaved processing
- Parallel processing
 - Processes C and D in figure below

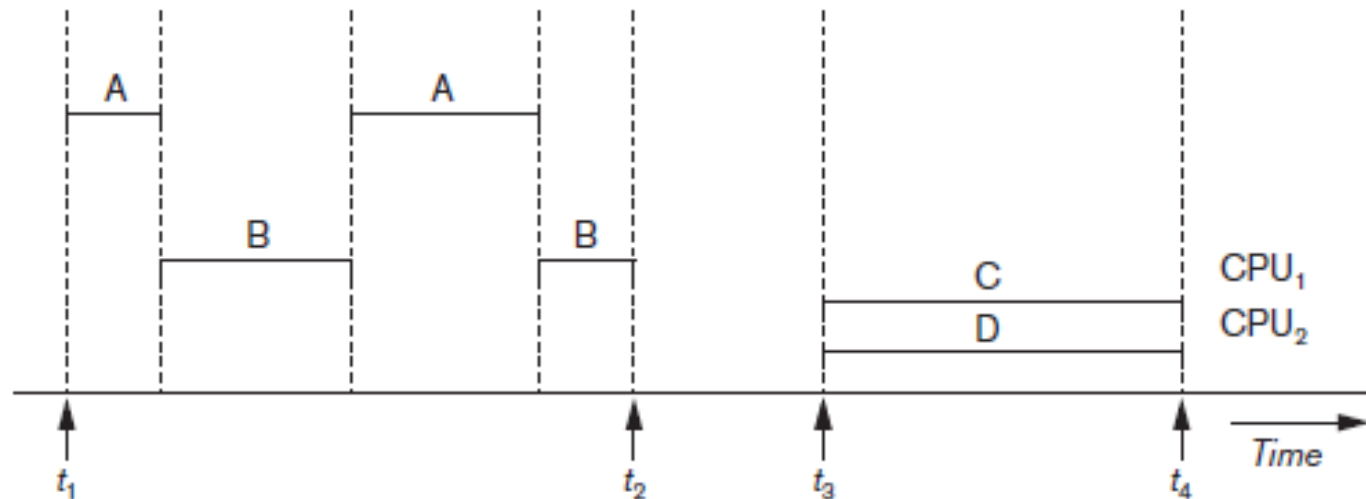


Figure 20.1 Interleaved processing versus parallel processing of concurrent transactions

Transactions

- Transaction:
 - an executing program that forms a logical unit of database processing
- Begin and end transaction statements
 - Specify transaction boundaries
- Read-only transaction
- Read-write transaction

Database Items

- Database represented as collection of named data items
- Size of a data item called its granularity
- Data item
 - Record
 - Disk block
 - Attribute value of a record
- Transaction processing concepts independent of item granularity

Read and Write Operations

- `read_item(X)`
 - Reads a database item named X into a program variable named X
 - Process includes
 - finding the address of the disk block
 - copying the whole block to a memory buffer
 - Copying the data item from memory buffer into program variable.
- `write_item(X)`
 - Writes the value of program variable X into the database item named X
 - Process includes
 - finding the address of the disk block,
 - copying the whole disk block to a memory buffer
 - Writing the data item into the disk block in memory buffer
 - storing the updated disk block back to disk

Read and Write Operations (cont'd.)

- Read set of a transaction
 - Set of all items read
- Write set of a transaction
 - Set of all items written

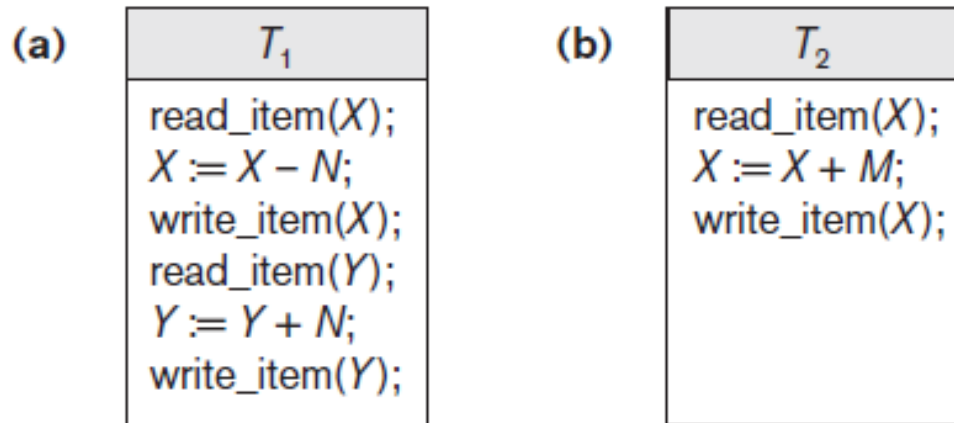


Figure 20.2 Two sample transactions (a) Transaction T_1 (b) Transaction T_2

DBMS Buffers

- DBMS will maintain several main memory data buffers in the database cache
- When buffers are occupied, a buffer replacement policy is used to choose which buffer will be replaced
 - Example policy: least recently used

Concurrency Control

- Transactions submitted by various users may execute concurrently
 - Access and update the same database items
 - Some form of concurrency control is needed
- The lost update problem
 - Occurs when two transactions that access the same database items have operations interleaved
 - Results in incorrect value of some database items

The Lost Update Problem

- Initially X is 90
- T1 transfers 5 seats from X to Y
- T2 reserves 4 seats in X
- Problem: T2 reads X before it gets updated by T1.

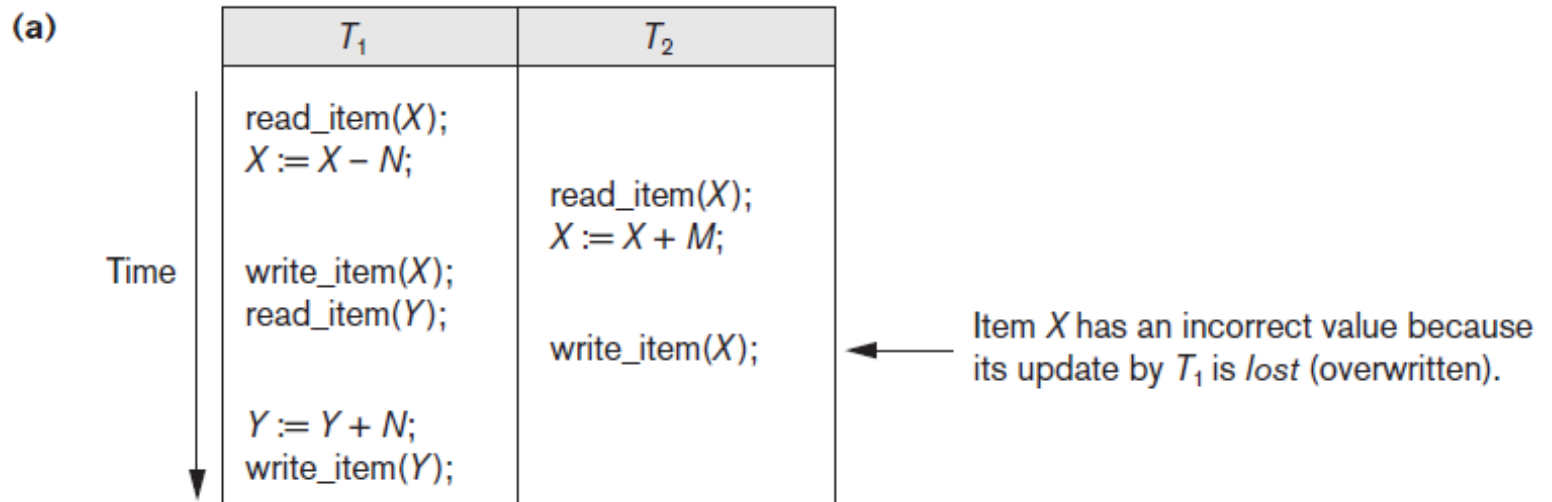


Figure 20.3 Some problems that occur when concurrent execution is uncontrolled (a) The lost update problem

The Temporary Update Problem

- T1 fails after updating X
- T2 reads X before T1 rolls back X

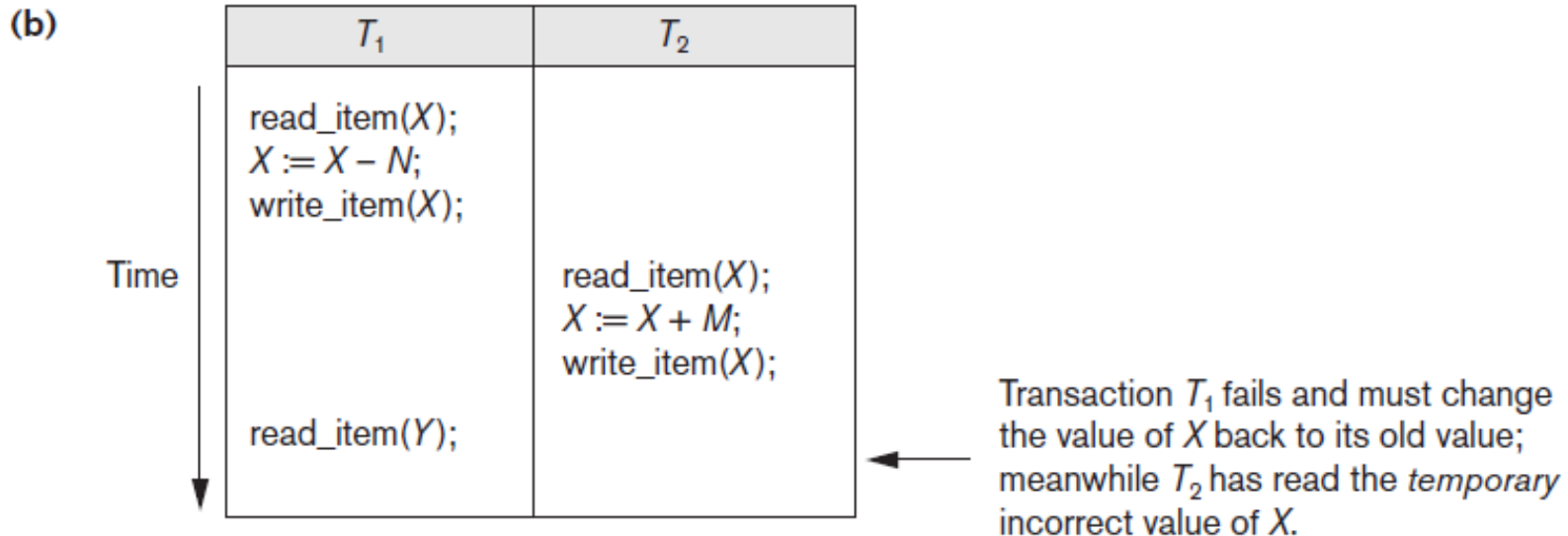


Figure 20.3 (cont'd.) Some problems that occur when concurrent execution is uncontrolled (b) The temporary update problem

The Incorrect Summary Problem

- T3 reads some values before update (Y) and some after update (X)

(c)

T_1	T_3
<pre>read_item(X); X := X - N; write_item(X); read_item(Y); Y := Y + N; write_item(Y);</pre>	<pre>sum := 0; read_item(A); sum := sum + A; ⋮ ⋮ ⋮ read_item(X); sum := sum + X; read_item(Y); sum := sum + Y;</pre>

← T_3 reads X after N is subtracted and reads Y before N is added; a wrong summary is the result (off by N).

Figure 20.3 (cont'd.) Some problems that occur when concurrent execution is uncontrolled (c) The incorrect summary problem

The Unrepeatable Read Problem

- Transaction T reads the same item twice
- Value is changed by another transaction T' between the two reads
- T receives different values for the two reads of the same item

Why Recovery is Needed

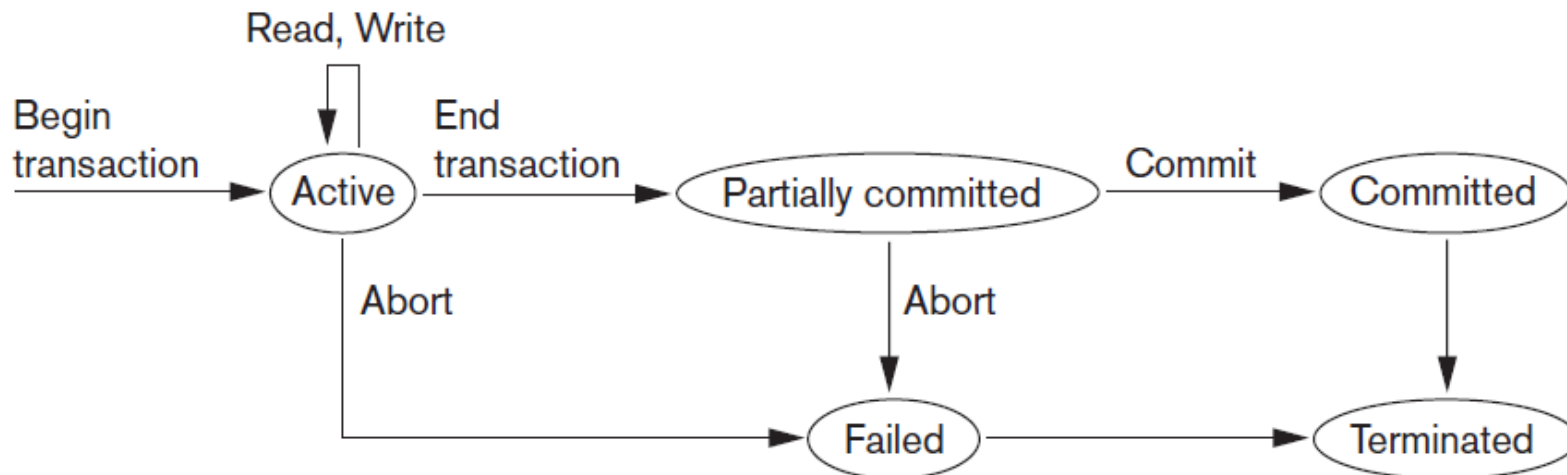
- Committed transaction
 - Effect recorded permanently in the database
- Aborted transaction
 - Should not affect the database
 - System must keep sufficient information to recover quickly from the failure
 - Disk failure or other catastrophes have long recovery times

Why Recovery is Needed (cont'd.)

- Types of transaction failures
 - Computer failure (system crash)
 - memory failure
 - Transaction or system error
 - Overflow or divide by zero
 - Local errors or exception conditions detected by the transaction
 - Errors due to unallowable operations
 - Concurrency control enforcement
 - Disk failure
 - Physical problems or catastrophes

20.2 Transaction and System Concepts

- System must keep track of when each transaction starts, terminates, commits, and/or aborts
 - BEGIN_TRANSACTION
 - READ or WRITE
 - END_TRANSACTION
 - COMMIT_TRANSACTION
 - ROLLBACK (or ABORT)



The System Log

- System log keeps track of transaction operations
- Sequential, append-only file
- Not affected by failure (except disk or catastrophic failure)
- Log buffer
 - Main memory buffer
 - When full, appended to end of log file on disk
- Log file is backed up periodically
- Undo and redo operations based on log possible

Log File Entries

- **[start_transaction, T].**
 - Indicates that transaction T has started execution.
- **[write_item, T , X , *old_value*, *new_value*].**
 - Indicates that transaction T has changed the value of database item X from *old_value* to *new_value*.
- **[read_item, T , X].**
 - Indicates that transaction T has read the value of database item X .
- **[commit, T].**
 - Indicates that transaction T has completed successfully and affirms that its effect can be committed (recorded permanently) to the database.
- **[abort, T].** Indicates that transaction T has been aborted.

Commit Point of a Transaction

- Occurs when all operations that access the database have completed successfully
 - And effect of operations recorded in the log
- Transaction writes a commit record into the log
 - If system failure occurs, can search for transactions with recorded start_transaction but no commit record
- Force-writing the log buffer to disk
 - Writing log buffer to disk before transaction reaches commit point

DBMS-Specific Buffer Replacement Policies

- Page replacement policy
 - Selects particular buffers to be replaced when all are full
- Domain separation (DS) method
 - Each domain handles one type of disk pages
 - Index pages
 - Data file pages
 - Log file pages
 - Number of available buffers for each domain is predetermined

DBMS-Specific Buffer Replacement Policies (cont'd.)

- Hot set method
 - Useful in queries that scan a set of pages repeatedly
 - Does not replace the hot set in the buffers until processing is completed
- The DBMIN method
 - Predetermines the pattern of page references for each algorithm for a particular type of database operation
 - Calculates locality set using query locality set model (QLSM)

20.3 Desirable Properties of Transactions

- ACID properties
 - Atomicity
 - Transaction performed in its entirety or not at all
 - Consistency preservation
 - Takes database from one consistent state to another
 - Isolation
 - Not interfered with by other transactions
 - Durability or permanency
 - Changes must persist in the database

Desirable Properties of Transactions (cont'd.)

- Levels of isolation
 - Level 0 isolation does not overwrite the dirty reads of higher-level transactions
 - Level 1 isolation has no lost updates
 - Level 2 isolation has no lost updates and no dirty reads
 - Level 3 (true) isolation has repeatable reads
 - In addition to level 2 properties
 - Snapshot isolation

20.4 Characterizing Schedules Based on Recoverability

- Schedule or history
 - Order of execution of operations from all transactions
 - Operations from different transactions can be interleaved in the schedule
- Total ordering of operations in a schedule
 - For any two operations in the schedule, they must occur in the same order as in the transactions.

Characterizing Schedules Based on Recoverability (cont'd.)

- Two operations in a schedule are said to be conflicting if:
 - Operations belong to different transactions and
 - Operations access the same item X and
 - At least one of the operations is a write_item(X)
- Two operations conflict if changing their order results in a different outcome
- Read-write conflict
- Write-write conflict

Characterizing Schedules Based on Recoverability (cont'd.)

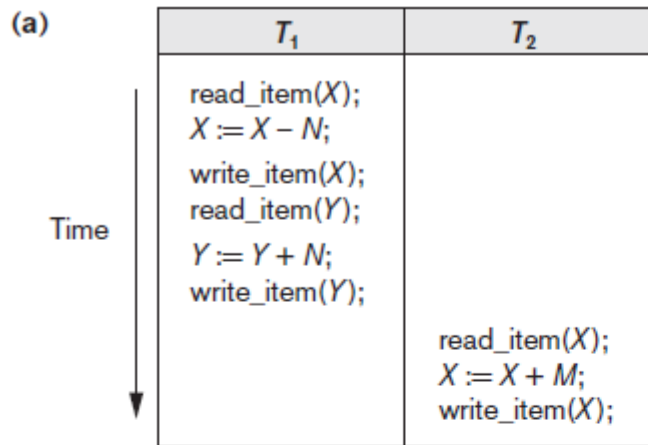
- Recoverable schedules
 - Recovery is possible
- Nonrecoverable schedules should not be permitted by the DBMS
- No committed transaction ever needs to be rolled back
- Cascading rollback may occur in some recoverable schedules
 - Uncommitted transaction may need to be rolled back

Characterizing Schedules Based on Recoverability (cont'd.)

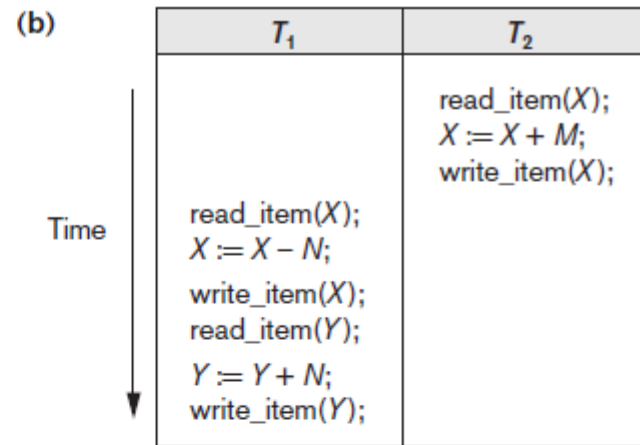
- Cascadeless schedule
 - Avoids cascading rollback
- Strict schedule
 - Transactions can neither read nor write an item X until the last transaction that wrote X has committed or aborted
 - Simpler recovery process
 - Restore the before image

20.5 Characterizing Schedules Based on Serializability

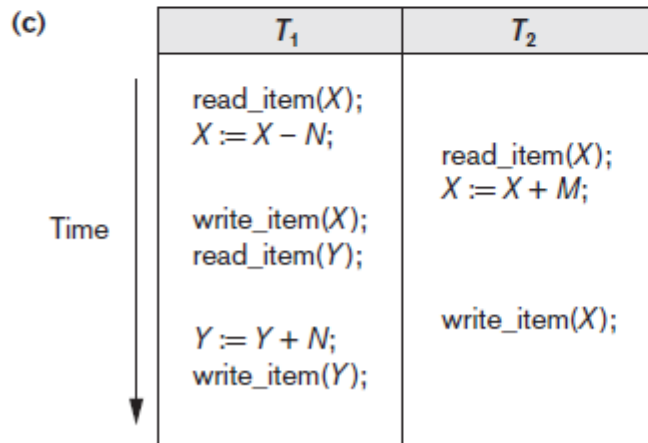
- **Serial Schedule:**
 - Places simultaneous transactions in series
 - Transaction T_1 before T_2 , or vice versa
 - Problem with serial schedules
 - Limit concurrency by prohibiting interleaving of operations
 - Unacceptable in practice
 - Solution: determine which schedules are equivalent to a serial schedule and allow those to occur



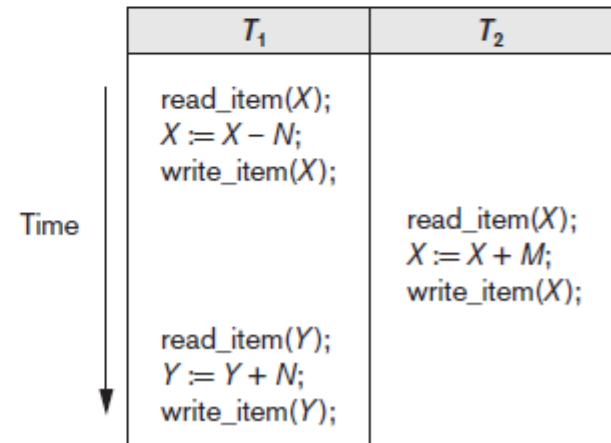
Schedule A



Schedule B



Schedule C



Schedule D

Figure 20.5 Examples of serial and nonserial schedules involving transactions T_1 and T_2 (a) Serial schedule A: T_1 followed by T_2 (b) Serial schedule B: T_2 followed by T_1 (c) Two nonserial schedules C and D with interleaving of operations

Characterizing Schedules Based on Serializability (cont'd.)

- Serializable schedules
 - Always considered to be correct when concurrent transactions are executing
- Serializable schedule of n transactions
 - Equivalent to some serial schedule of same n transactions

Characterizing Schedules Based on Serializability (cont'd.)

- Result equivalent schedules
 - Produce the same final state of the database
 - May be accidental
 - Cannot be used alone to define equivalence of schedules

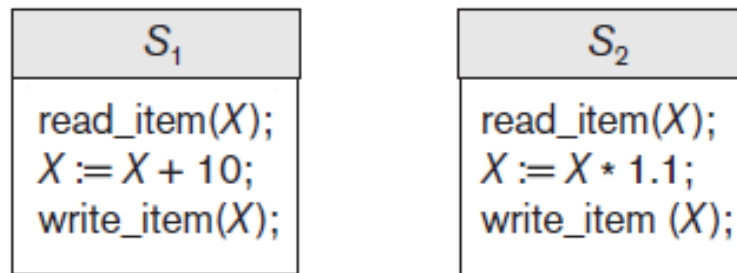


Figure 20.6 Two schedules that are result equivalent for the initial value of $X = 100$ but are not result equivalent in general

Characterizing Schedules Based on Serializability (cont'd.)

- Conflict equivalence
 - Relative order of any two conflicting operations is the same in both schedules
- Serializable schedules
 - Schedule S is serializable if it is conflict equivalent to some serial schedule S' .

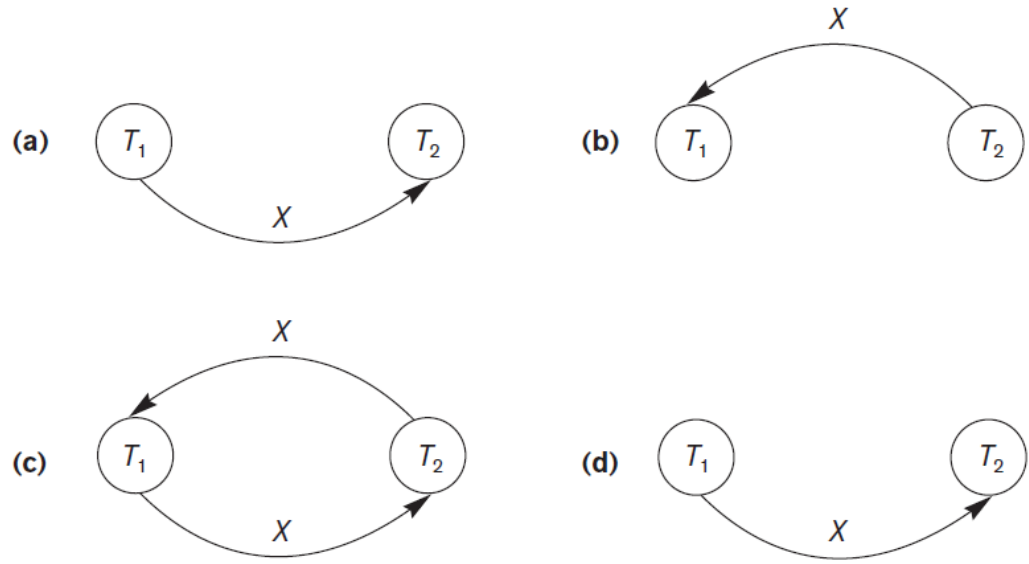
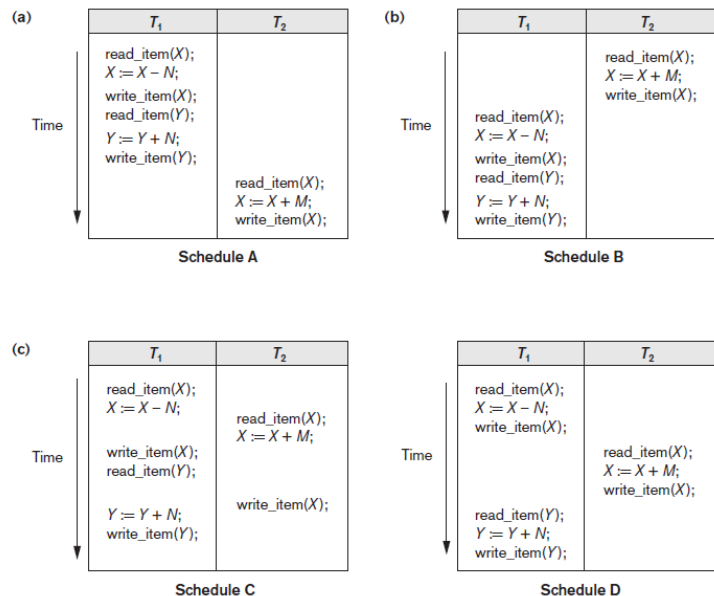
Characterizing Schedules Based on Serializability (cont'd.)

■ Testing for serializability of a schedule

1. For each transaction T_i participating in schedule S , create a node labeled T_i in the precedence graph.
2. For each case in S where T_j executes a `read_item(X)` after T_i executes a `write_item(X)`, create an edge $(T_i \rightarrow T_j)$ in the precedence graph.
3. For each case in S where T_j executes a `write_item(X)` after T_i executes a `read_item(X)`, create an edge $(T_i \rightarrow T_j)$ in the precedence graph.
4. For each case in S where T_j executes a `write_item(X)` after T_i executes a `write_item(X)`, create an edge $(T_i \rightarrow T_j)$ in the precedence graph.
5. The schedule S is serializable if and only if the precedence graph has no cycles.

Algorithm 20.1 Testing conflict serializability of a schedule S

Characterizing Schedules Based on Serializability (cont'd.)



1. For each transaction T_i participating in schedule S , create a node labeled T_i in the precedence graph.
2. For each case in S where T_j executes a `read_item(X)` after T_i executes a `write_item(X)`, create an edge $(T_i \rightarrow T_j)$ in the precedence graph.
3. For each case in S where T_j executes a `write_item(X)` after T_i executes a `read_item(X)`, create an edge $(T_i \rightarrow T_j)$ in the precedence graph.
4. For each case in S where T_j executes a `write_item(X)` after T_i executes a `write_item(X)`, create an edge $(T_i \rightarrow T_j)$ in the precedence graph.
5. The schedule S is serializable if and only if the precedence graph has no cycles.

How Serializability is Used for Concurrency Control

- Being serializable is different from being serial
- Serializable schedule gives benefit of concurrent execution
 - Without giving up any correctness
- Difficult to test for serializability in practice
 - Factors such as system load, time of transaction submission, and process priority affect ordering of operations
- DBMS enforces protocols
 - Set of rules to ensure serializability

View Equivalence and View Serializability

- View equivalence of two schedules
 - As long as each read operation of a transaction reads the result of the same write operation in both schedules, the write operations of each transaction must produce the same results
 - Read operations said to see the same view in both schedules
- View serializable schedule
 - View equivalent to a serial schedule

View Equivalence and View Serializability (cont'd.)

- Conflict serializability similar to view serializability if constrained write assumption (no blind writes) applies
- Unconstrained write assumption
 - Value written by an operation can be independent of its old value
- Debit-credit transactions
 - Less-stringent conditions than conflict serializability or view serializability

20.6 Transaction Support in SQL

- No explicit Begin_Transaction statement
- Every transaction must have an explicit end statement
 - COMMIT
 - ROLLBACK
- Access mode is
 - READ ONLY or
 - (default) READ WRITE
- Diagnostic area size option
 - Integer value (n) indicating number of conditions held simultaneously in the diagnostic area
 - Supply feedback (errors or exceptions) on the last “n” executed SQL statements.

Transaction Support in SQL (cont'd.)

- Isolation level option
 - Dirty read
 - Nonrepeatable read
 - Phantoms

Isolation Level	Type of Violation		
	Dirty Read	Nonrepeatable Read	Phantom
READ UNCOMMITTED	Yes	Yes	Yes
READ COMMITTED	No	Yes	Yes
REPEATABLE READ	No	No	Yes
SERIALIZABLE	No	No	No

Table 20.1 Possible violations based on isolation levels as defined in SQL

Transaction Support in SQL (cont'd.)

- Snapshot isolation
 - Used in some commercial DBMSs
 - Transaction sees data items that it reads based on the committed values of the items in the database snapshot when transaction starts
 - Ensures phantom record problem will not occur

20.7 Summary

- Single and multiuser database transactions
- Uncontrolled execution of concurrent transactions
- System log
- Failure recovery
- Committed transaction
- Schedule (history) defines execution sequence
 - Schedule recoverability
 - Schedule equivalence
- Serializability of schedules